

12 (a)	strong uniform (magnetic) field	M1	
	<i>either</i> aligns nuclei		
	<i>or</i> gives rise to Larmor/resonant frequency <u>in r.f. region</u>	A1	
	non-uniform (magnetic) field	M1	
	<i>either</i> enables nuclei to be located		
	<i>or</i> changes the Larmor/resonant frequency	A1	[4]
(b) (i)	difference in flux density = $2.0 \times 10^{-2} \times 3.0 \times 10^{-3} = 6.0 \times 10^{-5} \text{ T}$	A1	[1]
(ii)	$\Delta f = 2 \times c \times \Delta B$	C1	
	$= 2 \times 1.34 \times 10^8 \times 6.0 \times 10^{-5}$		
	$= 1.6 \times 10^4 \text{ Hz}$	A1	[2]

Page 7	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2012	9702	41

13 (a) (i) no interference (between signals) near boundaries (of cells) B1 [1]